

Identity Recognition for Multiple Agents in An Alternative-Paths Workflow: A Hidden Markov Approach

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Abstract—In various business, industrial, and computational domains, a supervised common environment often provides shared resources for executing a collection of supported tasks. Multiple agents, acting as legitimate users, may repeatedly check in, select a task, and follow the designated workflow of the task to completion. Essentially, each agent continuously engages in a common alternative-paths workflow, where the workflows of individual tasks define the available alternative paths. Computer systems and interconnected sensor networks are widely used to collect sequences of observations from these agent processes, which serve as their behavioral signatures. For identity recognition in such an environment, a key computational challenge is to analyze the collected signatures and identify the most likely agents responsible for generating them. An anomaly is detected if the claimed owner of a process is not among the most likely identified agents. This capability enables the system to monitor agent behavior and detect potential security breaches. In this paper, we employ hidden Markov models (HMMs) to model the behavioral dynamics of agents engaging in alternative-paths workflows. Differences in agent behavior are characterized by distinct parameter values associated with the HMMs. We present a computational framework for identity recognition in this context and conduct experiments in the domain of text input using a circular keyboard. Our empirical results demonstrate the feasibility of this framework and its effectiveness for identity recognition in this domain.

Index Terms—Machine learning, pattern recognition, probabilistic reasoning, workflows, and processes

I. INTRODUCTION AND LITERATURE REVIEW

Computer systems and interconnected sensor networks are widely used to monitor processes across industrial, business, and human-machine interaction domains. [1]–[6]. Such a process often arises when an agent engages in an underlying workflow consisting of interconnected operational stages through which a task progresses from initiation to completion [4], [7]–[9]. For example, the human-machine interaction process of a user typing a word follows a sequential workflow of interconnected stages arranged in a linear order, as shown in Fig. 1a. The initial stage I represents the

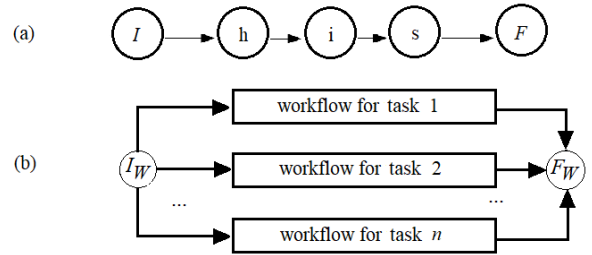


Fig. 1. (a) A sequential workflow w for the task of typing the word “his”. (b) An alternative-paths workflow $alt(W)$ for executing a task selected from a collection W of n tasks by an agent.

cognitive readiness to start typing, while the final stage F marks the completion of the process by entering a delimiter. Each intermediate stage corresponds to pressing a specific key on the keyboard to generate the corresponding character in the word. The alternative-paths workflow [10] in Fig. 1b models the generic process in which an agent p executes a task selected from a collection W of n tasks, represented by the alternative paths in the diagram. This is also known as the exclusive-choice workflow pattern, as described in [8]. The initial stage I_W determines which task to proceed with, while the final stage F_W concludes the task, marking the end of the process. If the tasks along the alternative paths represent sequential workflows for typing individual words from a vocabulary W of n words, then the workflow in Fig. 1b models the process of selecting and typing a word from this vocabulary. Repeatedly executing the alternative-paths workflow in Fig. 1b in this context models an agent repeatedly typing words selected from this vocabulary.

Observations from the operational stages of the underlying workflow can be collected over time for pattern recognition, security assessment, anomaly detection, and other decision-making applications [1]–[6]. By analyzing sequences of observations, process management systems or adversaries may infer the identity and intentions of the agent engaging

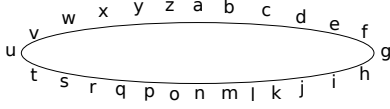


Fig. 2. A circular virtual-reality keyboard for people with special needs.

in the underlying workflow [1], [4], [5], [10], [11]. Previous research in [1] investigated security issues related to the login process of entering a password, as represented in the workflow in Fig. 1a. The Secure Shell (SSH) protocol is commonly used to establish a secure, encrypted channel between a user and a remote host. However, eavesdroppers can infer the user's inter-keystroke timings from packet arrival times over the Internet. As shown in [1], a hidden Markov model [12], [13] can represent the probabilistic relationship between inter-keystroke time latencies and possible character sequences entered. This enables adversaries to predict the most likely passwords based on observed inter-keystroke time latencies, potentially compromising SSH security.

Another study in [4] investigated automatic text recognition for individuals with special needs using a circular virtual reality keyboard, as depicted in Fig. 2 [14]–[18]. When a user repeatedly types words selected from a given vocabulary, they act as an agent engaging in the alternative-paths workflow in Fig. 1b. Each pass through the workflow produces a sequence of characters observed, requiring the virtual reality system to infer the intended word (one of the tasks on the alternative paths). As shown in [4], hidden Markov models can represent the probabilistic relationships governing transitions between operational stages in the workflow and the uncertainty of observations at each stage. The research in [4] describes a computational framework for identifying the most likely tasks along the alternative paths (i.e., the most likely words in the vocabulary) that correspond to observed sequences (i.e., noisy output strings). Empirical results in [4] demonstrated the effectiveness of automatic text recognition from noisy output strings generated by a known agent.

In this paper, we investigate the identity recognition problem in an environment where multiple agents repeatedly work on a collection of routine tasks $\mathcal{W}$, with each agent selecting and completing one of the tasks in $\mathcal{W}$ one at a time. Each of these routine tasks has its own workflow and the collection of tasks together defines an alternative-paths workflow $alt(\mathcal{W})$ as shown in Fig. 1b. In other words, each agent is repeatedly engaging in an alternative-paths workflow where the workflows on the alternative paths are the workflows of the routine tasks. When an agent engages in the specific workflow of a routine task, (i) the probabilistic relationships governing transitions between operational stages in the workflow and (ii) the uncertainty of observations at each stage is represented as a hidden Markov model in terms of several characteristic parameter values as

depicted in [4]. For each specific routine task, the hidden Markov models associated with the agents share the same topological state-transition structure since they are all based on the same underlying workflow. As proposed in [4], the differences of agents are modeled in terms of the characteristic parameter values regarding the transition probabilities among the stages and the observational probabilities over the stages. The sequences of observations resulting from an unverified agent repeatedly engaging in an alternative-paths workflow are essentially behavioral signatures rendered by the agent. For identity recognition in such an environment, the computational problem is to examine these signatures and identify the most likely agents that may generate these signatures. An anomaly is detected if the claimed owner of a process is not among the most likely agents identified. This would enable the system to monitor the behavior of engaging agents and identify potential security breaches.

In the remainder of the paper, we present a computational framework of identity recognition for agents in an alternative-paths workflow. We conduct an empirical study on the effectiveness of the framework in the context of text entry by people with special needs using a circular virtual-reality keyboard [4]. Our empirical results show the feasibility of identity recognition based on noisy output strings generated by people with special needs. Section II summarizes the related terms, definitions, and notations regarding hidden Markov processes, alternative-paths workflows, and the parameterizations of transition probabilities and observation probabilities following the convention of [4]. Section III depicts the computational framework of identity recognition. In Section IV, we describe the settings of and results from an empirical study of identity recognition in the context of text entry by people with special needs using a circular virtual-reality keyboard [4], [18], demonstrating the feasibility of the identity recognition framework. Finally, we summarize the empirical findings in Section V and provides the conclusions in Section V.

II. THE IDENTITY RECOGNITION PROBLEM

In this section, we focus on the application of hidden Markov processes for identity recognition and security assessment. A hidden Markov process [4] is a hidden Markov model [4], [12] representing the dynamics of a workflow in which (i) one, none, or multiple observations may be obtained after entering a stage each time, and (ii) the progress can only be indirectly monitored through the observations collected over time, without exact knowledge of the alignment between the observations and the source stages. In this hidden Markov model, (i) the hidden states are the operational stages that yield the observations, (ii) exactly one observation is obtained after entering a state each time, and (iii) in any current state other than the final stage, it may transition to itself or another state corresponding to a stage reachable from the current state

(as a stage) in the workflow. The uncertainty associated with transitions and observations over stages is modeled by transition probabilities and observation probabilities over the stages. In the following, we formally define all the related terms and the computational problem of identity recognition for hidden Markov processes with a common alternative-paths workflow.

Definition 1 (Workflows). A workflow $w = \langle S, \prec, \mathcal{I}, \mathcal{F} \rangle$ is a collection of operational stages S connected by a set of directed edges $\prec$ over S , forming a directed graph $G = \langle S, \prec \rangle$. Each directed edge in $\prec$ indicates a possible transition from one stage to another stage in the workflow. An initial stage $\mathcal{I}$ and a final stage $\mathcal{F}$ in S are the source and the sink of G respective, designating the beginning and the end of the workflow.

Definition 2 (A hidden Markov process modeling an agent in a workflow). A hidden Markov process HMM_w^p modeling the process of an agent p completing a workflow $w = \langle S, \prec, \mathcal{I}, \mathcal{F} \rangle$ for one time is a process that can be described as a hidden Markov model $HMM_w^p = \langle S, O, \pi, p_{tran}, p_{obs} \rangle$ where (i) the set of operational stages S in the workflow is the set of hidden states, not directly observable in general, (ii) O is the set of possible observations obtained from individual stages, (iii) the process starts with the initial stage $\mathcal{I}$, (iv) the process ends when it enters the final stage $\mathcal{F}$, (v) p_{tran} represents the state transition probabilities where $p_{tran}(s_j | s_i)$ is the transition probability to stage s_j as the stage associated with the next observation given stage s_i as the stage associated with the current observation, and (vi) p_{obs} represents the state observation probabilities where $p_{obs}(o_j | s_i)$ is the probability of observing $o_j \in O$ given stage s_i as the stage associated with the current observation.

Example (Hidden Markov processes): Consider the hidden Markov process HMM_w^p modeling an agent p engaging the workflow w in Fig. 1a to type the word “his” using the circular keyboard in Fig. 2. Fig. 3 demonstrates the transition probabilities of HMM_w^p while Table I shows part of the observation probabilities of HMM_w^p as described in the probabilistic modeling of the circular keyboard in [4].

Definition 3 (A signature of an agent in a workflow). A hidden Markov signature of an agent p engaging a workflow w is the sequence of observations $\vec{O}$ obtained from the hidden Markov process HMM_w^p from the point HMM_w^p starts from the initial stage to the point HMM_w^p ends in the final stage.

Definition 4 (Alternative-Paths workflows). Given a collection of workflows $\mathcal{W} = \{w_1, w_2, \dots, w_n\}$, the alternative-paths workflow $alt(\mathcal{W}) = \langle S_{\mathcal{W}}, \prec_{\mathcal{W}}, \mathcal{I}_{\mathcal{W}}, \mathcal{F}_{\mathcal{W}} \rangle$

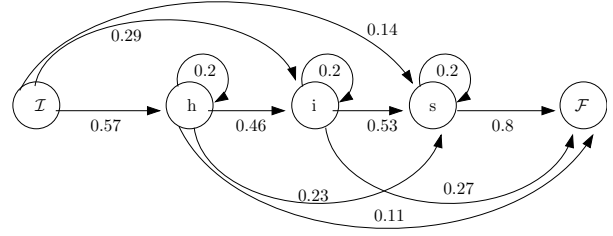


Fig. 3. Transition probabilities of the hidden Markov process HMM_w^p modeling an agent p (with $deg_{tran} = 2$ and $p_{iter} = 0.2$) engaging the workflow w in Fig. 1a.

TABLE I
OBSERVATION PROBABILITIES OF THE HIDDEN MARKOV PROCESS HMM_w^p FOR AN AGENT p (WITH $p_{mode} = 0.3$ AND $deg_{obs} = 2$) ENGAGING THE WORKFLOW w IN FIG. 1A USING THE KEYBOARD IN FIG. 2.

Stage	Possible observations (i.e. keys possibly touched)					
	a	...	h	i	j	...
h	0.00273		0.3	0.17503	0.08752	0.00017
i	0.00137		0.17503	0.3	0.17503	0.00034
s	0.00137		0.00017	0.00034	0.00068	0.3

is a directed graph $G_{\mathcal{W}}$ where (i) the workflows w_i 's in $\mathcal{W}$ are disjoint subgraphs in $G_{\mathcal{W}}$, (ii) the initial stage $\mathcal{I}_{\mathcal{W}}$ is the source that only leads to the initial stages of w_i 's, and (iii) the final stages of w_i 's only connects to the final stage $\mathcal{F}_{\mathcal{W}}$ as the sink. Any path from $\mathcal{I}_{\mathcal{W}}$ to $\mathcal{F}_{\mathcal{W}}$ in the alternative-paths workflow $alt(\mathcal{W})$ thus passes through exactly one of the workflows w_i 's.

Alternative-paths workflows as the standard form of workflows: Every workflow w can be equivalently viewed as an alternative-paths workflow $alt(\{w\})$ with only one alternative path. Therefore, without loss of generality, we formulate the identity recognition problem and the computational framework in the context of alternative-paths workflows in the remainder of the paper.

Definition 5 (Identity recognition for agents in an alternative-paths workflow). Given (i) a shared alternative-paths workflow $alt(\mathcal{W})$ composed of a set of n workflows $\mathcal{W} = \{w_1, \dots, w_n\}$ on its alternative paths, (ii) a set of l agents $\mathcal{P} = \{p_1, \dots, p_l\}$ with each agent p in $\mathcal{P}$ repeatedly engaging in the workflows in $\mathcal{W}$ one at a time, (iii) for each agent p in $\mathcal{P}$ and each workflow w in $\mathcal{W}$, the probability information of the hidden Markov process HMM_w^p , and (iv) a set of m signatures $\vec{O} = \{\vec{O}_1, \dots, \vec{O}_m\}$ generated by hidden Markov processes HMM_w^p associated with an unidentified agent p and unidentified workflows w , the computational task of identity recognition is to identify (for a given constant c) the top c most likely agents in $\mathcal{P}$ who could have generated the given signatures.

III. COMPUTATIONAL FRAMEWORK FOR IDENTITY RECOGNITION

In the following, we describe the computational approach for solving the identity recognition problem. We first derive the mathematical formulation for identifying the most likely agent given multiple observed signatures. We then describe how we can apply the forward algorithm for hidden Markov models as the core computational component to determine the most likely agents that may generate these observed signatures.

Maximum Likelihood formulation for identifying the most likely agents: Let P be the random variable denoting the identity of the unknown agent (i.e. the underlying characteristic parameter setting of the agent) executing the task. To infer the most likely agent out of a set of l possible agents $\mathcal{P} = \{p_1, p_2, \dots, p_l\}$ given a collection of m observed signatures $\vec{O} = \{\vec{O}_1, \dots, \vec{O}_m\}$, we can apply the maximum likelihood principle based on the following formulation

$$\begin{aligned} & \arg \max_{p_i \in \mathcal{P}} Pr(\vec{O}_1, \dots, \vec{O}_m | P = p_i) \\ &= \arg \max_{p_i \in \mathcal{P}} \left(\prod_{1 \leq k \leq m} Pr(\vec{O}_k | P = p_i) \right) \\ &= \arg \max_{p_i \in \mathcal{P}} \left(\log \prod_{1 \leq k \leq m} Pr(\vec{O}_k | P = p_i) \right) \\ &= \arg \max_{p_i \in \mathcal{P}} \left(\sum_{1 \leq k \leq m} \log Pr(\vec{O}_k | P = p_i) \right). \end{aligned} \quad (1)$$

Computational elements for the maximum Likelihood formulation: Equation 1 above requires the calculation of $Pr(\vec{O}_k | P = p_i)$, the probability of an observed signature $\vec{O}_k$ is the result of an agent $P = p_i$ executing some unknown task in the alternative-paths workflow. Let W be the random variables denoting the unknown task executed by the agent. Given an observed signature $\vec{O}_k$, we can formulate the probability that $\vec{O}_k$ is the result of agent P executing some unknown task in $\mathcal{W} = \{w_1, w_2, \dots, w_n\}$ as

$$\begin{aligned} & Pr(\vec{O}_k | P) \\ &= \sum_{w_j \in \mathcal{W}} Pr(\vec{O}_k, W = w_j | P) \\ &= \sum_{w_j \in \mathcal{W}} Pr(\vec{O}_k | W = w_j, P) * Pr(W = w_j | P) \\ &= \sum_{w_j \in \mathcal{W}} Pr(\vec{O}_k | W = w_j, P) * 1/n \\ &= 1/n * \sum_{w_j \in \mathcal{W}} Pr(\vec{O}_k | W = w_j, P). \end{aligned} \quad (2)$$

Assumption 1 (Modeling $Pr(W|P)$ as a uniform distribution). In the formulation above, we assume for every agent each of the n workflows in $\mathcal{W}$ is equally likely and thus $Pr(W = w_j | P = p_i) = 1/n$. By assuming $Pr(W|P)$ follows a uniform distribution, we model the situation where

no prior information is available to distinguish agents based on how frequently these agents engage with specific workflows in $\mathcal{W}$. Combining Equations 1 and 2 together under Assumption 1, we can determine the most like agent based on the following formula:

$$\begin{aligned} & \arg \max_{p_i \in \mathcal{P}} Pr(\vec{O}_1, \dots, \vec{O}_m | P = p_i) \\ &= \arg \max_{i=1}^l \sum_{k=1}^m \log \sum_{j=1}^n Pr(\vec{O}_k | W = w_j, P = p_i). \end{aligned} \quad (3)$$

Therefore, to determine the most likely agents, we need to determine $Pr(\vec{O}_k | W = w_j, P = p_i)$ for every given observed signature $\vec{O}_k$, task w_j , and agent p_i , which can be accomplished by using the well-known forward algorithm for hidden Markov models [12]. This leads to Algorithm 1 below for identifying the most likely agents based on the observed signatures.

Algorithm 1: Identification of the most likely agents.

- Input: A set $\mathcal{P}$ of l agents, a set $\vec{O}$ of m signatures, a set $\mathcal{W}$ of n tasks, a constant c , and the probability information of hidden Markov models HMM_w^p for every p in $\mathcal{P}$ and every w in $\mathcal{W}$.
- Output: The top c most likely agents in $\mathcal{P}$ who could have generated the collection of signatures $\vec{O}$.
- Step 1: For each of the l agents p_i in $\mathcal{P}$, each of the m signatures $\vec{O}_k$ in $\vec{O}$, and each of the n tasks w_j in $\mathcal{W}$, apply the forward algorithm to $HMM_{w_j}^{p_i}$ to determine $Pr(\vec{O}_k | W = w_j, P = p_i)$, the probability of observing the signature O_k from $HMM_{w_j}^{p_i}$.
- Step 2: For each of the l agents p_i in $\mathcal{P}$, determine $\sum_{k=1}^m \log \sum_{j=1}^n Pr(\vec{O}_k | W = w_j, P = p_i)$ as the probabilistic score for p_i as depicted in Equation 3.
- Step 3: Report the agents associated with the top c scores calculated above according to Equation 3.

Proposition 1 (Identity recognition based on Algorithm 1). Given a signature $\vec{O}_k$ and the probability information of the hidden Markov model $HMM_{w_j}^{p_i}$ regarding agent p_i engaging in workflow w_j , the forward algorithm for hidden Markov models can determine $Pr(\vec{O}_k | W = w_j, P = p_i)$, the probability of observing $\vec{O}_k$ as the result of process modeled by $HMM_{w_j}^{p_i}$ in $O(|O_k| * |w_j|^2)$ time [12], where $|O_k|$ is the length of the signature O_k and $|w_j|$ is the number of stages in the workflow w_j . Consequently, Algorithm 1 can correctly identify the most likely agents generating the observed signatures according to Equation 3.

IV. AN EMPIRICAL STUDY ON IDENTITY RECOGNITION GIVEN SIGNATURES GENERATED FROM A VIRTUAL KEYBOARD

We conducted an empirical study on identity recognition in the context of a virtual reality environment where people

TABLE II
ACCURACY OF IDENTITY RECOGNITION: PERCENTAGE OF TIME TRUE AGENTS CORRECTLY IDENTIFIED AMONG THE TOP k MOST LIKELY

Vocabulary Set: RNT (94 unique words)				
# of signatures	10	30	50	Average
Top 1	43.2%	39.5%	37.0%	39.9%
Top 3	79.0%	85.2%	77.8%	80.7%
Top 5	90.1%	93.8%	91.4%	91.4%
Vocabulary Set: QBF (8 unique words)				
# of signatures	10	30	50	Average
Top 1	44.4%	27.2%	35.8%	35.8%
Top 3	81.5%	67.9%	80.2%	76.5%
Top 5	92.6%	90.1%	93.8%	92.2%
Vocabulary Set: JUMPS (1 word)				
# of signatures	10	30	50	Average
Top 1	33.3%	45.7%	49.4%	42.8%
Top 3	74.1%	85.2%	85.2%	81.5%
Top 5	90.1%	95.1%	93.8%	93.0%

generate text output using a circular virtual keyboard [4], [18] as depicted in Figure 2. Consider (i) a vocabulary of n words $\mathcal{W} = \{w_1, \dots, w_n\}$, (ii) a group of l agents $\mathcal{P} = \{p_1, \dots, p_l\}$ who can check in the virtual environment and use the circular keyboard to type words selected from $\mathcal{W}$, and (iii) a set of m text strings as signatures $\vec{\mathcal{O}} = \{\vec{\mathcal{O}}_1, \dots, \vec{\mathcal{O}}_m\}$ resulting from an unidentified agent in $\mathcal{P}$ typing m unidentified words in $\mathcal{W}$ using the circular keyboard. In other words, each agent is repeatedly engaging in an alternative-paths workflow $alt(\mathcal{W})$ where the workflows on the alternative paths are simply the workflows for typing the individual words in $\mathcal{W}$ as shown in Figure 1. In this context, our identity recognition task is to determine the top c most likely agents in $\mathcal{P}$ based on the m text signatures collected. Note that we do not know the underlying words corresponding to these m text signatures in $\vec{\mathcal{O}}$. All we know is that they are the results of one of the agents in $\mathcal{P}$ typing m words selected from $\mathcal{W}$.

In the remainder of this section, we (i) first describe the vocabulary sets used in the experiments and the four characteristic parameters for modeling the behavioral differences of agents, (ii) then present the parameter settings and the experimental framework of the empirical study on identity recognition, and (iii) provide empirical results on the accuracy of identity recognition across various settings to demonstrate the effectiveness of the framework. Table II summarizes the accuracy of identity recognition on average under various settings in the context of three different vocabulary sets.

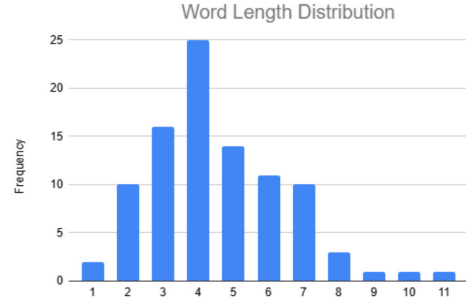


Fig. 4. The distribution of word length (the number of stages other than $\mathcal{I}$ and $\mathcal{F}$ in the workflow for typing the word) in the RNT vocabulary

Three vocabulary sets: We use three different vocabulary sets in the experiments: (i) RNT: the famous poem "The Road Not Taken" by Robert Frost, which contains 94 unique words, (ii) QBF: a short sentence, "The quick brown fox jumps over the lazy dog," which contains 8 unique words, and (iii) JUMPS: the single word "jumps." See Figure 4 about the distribution of word length (i.e. the number of stages other than $\mathcal{I}$ and $\mathcal{F}$ in the workflow for typing the word) in the RNT vocabulary.

Four characteristic parameters for modeling the differences of agents: As described above in this section, all the agents in $\mathcal{P}$ engage in the same alternative-paths workflow $alt(\mathcal{W})$. However, they are different in terms of the details of their typing behaviors. We follow the framework proposed in [4], and model the behavioral differences of agents in terms of the following four characteristic parameter values regarding the transition probabilities among the stages and the observational probabilities over the stages in the workflows.

(i) Iterative locality, p_{iter} : The parameter p_{iter} represents the probability of reiterating one more iteration in the same stage of the workflow after finishing one iteration of operation in that stage. In our experimental context, it is the probability that the agent types the same letter in the word again instead of moving on to the next.

(ii) Transition metric locality, deg_{tran} : Given the stage that generates the current observable output, there is uncertainty regarding what the stage that generates the next observable output could be. The parameter deg_{tran} represents the base for an exponential decrease in transition probability and we model the transition probability as deg_{tran}^{-d} multiplied by a scaling constant for a transition with a transition distance d . The distance between two stages in the workflow is measured in terms of the distance in the workflow (as a directed graph). In our experimental context, the larger the value of deg_{tran} is the less likely (exponentially) that letters in a word may be skipped without outputs.

(iii) Observation modal locality, p_{mode} : We use p_{mode} to

parameterize the probability of observing the most likely observation output (i.e. the characteristic observation item) associated with the stage. In our experimental context, p_{mode} is the probability that the agent accurately presses the target letter (of the current stage) to correctly produce the intended output observed.

(iv) **Observation metric locality**, deg_{obs} : There is uncertainty regarding the actual output observed when the agent given that the agent is in a specific stage of the workflow. In our experimental context, the circular keyboard defines a distance metric over the twenty six letters on the keyboard, and we can measure the distance between the target letter (of the current stage) and the actual output letter observed. The parameter deg_{tran} represents the base for an exponential decrease in observation probability given the distance of the target letter and the actual letter observed, and we model the observation probability of the item as deg_{obs}^{-d} multiplied by a scaling constant. The larger the value of deg_{obs} is, the less likely the output character observed is far away from the target letter on the keyboard.

Parameter settings of agents in the empirical study: For all the stages of the workflows (other than the starting and terminating stages $\mathcal{I}$ and $\mathcal{F}$), we assign one of three possible parameter values (as show below) to each of the parameters p_{iter} , deg_{tran} , p_{mode} , and deg_{obs} , i.e.

- p_{iter} , iterative locality: 0.7, 0.4, or 0.1,
- deg_{tran} , transition metric locality: 4, 3, or 2,
- p_{mode} , observation modal locality: 0.9, 0.6, or 0.3, and
- deg_{obs} , observation metric locality: 3, 2, or 1.5.

All together, we have 81 different parameter settings in the parameter space. We utilize these parameter settings to represent 81 agents with different tendencies in their typing behavior. In other words, each agent is characterized by exactly one of the 81 parameter settings.

Generating the collections of signatures as test cases for the experiments: We implement a program for simulating the operation of the hidden Markov model HMM_w^p regarding an agent p typing a given word w . Given (i) the characteristic parameter setting of the agent p and (ii) a vocabulary set $\mathcal{W}$, the program can also serve as a generative model for probabilistically producing signatures as the result of agent p typing randomly selected words from $\mathcal{W}$. To generate test cases, for each of the three vocabulary sets above, we have each of the eighty one agents randomly pick and type 10, 30, and 50 words from the vocabulary set accordingly, leading to three collections of 10, 30, and 50 signatures respectively. Therefore, altogether we have 729 (i.e. $3 \times 81 \times 3$) collections of signatures as our test cases. For each of these collections of signatures, we know the identity of the agent generating the collection of signatures.

The experiments on the accuracy of identity recognition: Given the three possible vocabulary sets and the three possible sizes of signature collections, there are 9 specific combinations regarding the source vocabulary set and the number of signatures in the collection. For each of these 9 specific combinations, we have 81 signature collections and for each collection we know the identity of the agent generating the signatures. For each of these 9 specific combinations, we apply Algorithm 1 to identify the top 5 most likely agents for each of corresponding 81 signature collections. We then determine the percentage of time, the true author (agent) is correctly identified as one of the top 1, 3, or 5 most likely agents.

The empirical results on the accuracy of identity recognition: Tables II shows the accuracy of identity recognition for each specific combination of the vocabulary set used and the number of signatures in the collection for identity recognition.

The experiments on the identification error in the parameter space: For each signature collection, we also quantitatively measured the identification error $error(x, \hat{x})$ (as defined in Definition 6 below) between the parameter setting $(p_{iter}^x, p_{mode}^x, deg_{tran}^x, \text{ and } deg_{obs}^x)$ of the agent x that generates the collection and the parameter setting $(p_{iter}^{\hat{x}}, p_{mode}^{\hat{x}}, deg_{tran}^{\hat{x}}, \text{ and } deg_{obs}^{\hat{x}})$ of the most likely agent $\hat{x}$ identified by Algorithm 1.

Definition 6 (Identification error $error(x, \hat{x})$). We define the identification error between the true agent x and the most likely agent $\hat{x}$ identified according to the following normalized Manhattan distance in the parameter space

$$error(x, \hat{x}) = \frac{|p_{iter}^x - p_{iter}^{\hat{x}}|/0.6 + |deg_{tran}^x - deg_{tran}^{\hat{x}}|/2 + |p_{mode}^x - p_{mode}^{\hat{x}}|/0.6 + |deg_{obs}^x - deg_{obs}^{\hat{x}}|/1.5}{4}. \quad (4)$$

Note that for each of the four parameters we normalize the difference of the two parameter values involved in the formula according to the range of possible parameter values used in our experiments. Therefore, the normalized Manhattan distance $error(x, \hat{x})$ defined in Equation 4 above has the following properties.

Proposition 2 (The range of identification error as a Manhattan distance). In the parameter space in our experiments, the normalized Manhattan distance $error(x, \hat{x})$ between any two points x and $\hat{x}$ (i.e. two parameter settings modeling two agents) is no more than 4. The expected normalized Manhattan distance between two randomly selected points in the parameter space is $\frac{3}{4}$. When the true agent is correctly identified as the most likely agent, the Manhattan distance is zero. Otherwise, the

TABLE III
MEASURING IDENTIFICATION ERROR IN MANHATTAN DISTANCE
(OBSERVATION METRIC LOCALITY $deg_{obs} = 3$)

Transition metric locality: $deg_{tran} = 4$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.36 (0.56)	0.25 (0.33)	0.33 (0.33)
0.4	0.44 (0.72)	0.19 (0.17)	0.08 (0.33)
0.1	0.53 (0.17)	0.47 (0.17)	0.36 (0.17)
Transition metric locality: $deg_{tran} = 3$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.28 (0.00)	0.36 (0.33)	0.42 (0.22)
0.4	0.36 (0.17)	0.44 (0.17)	0.44 (0.39)
0.1	0.53 (0.50)	0.58 (0.72)	0.61 (0.72)
Transition metric locality: $deg_{tran} = 2$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.20 (0.56)	0.41 (0.17)	0.00 (0.33)
0.4	0.20 (0.33)	0.08 (0.56)	0.48 (0.33)
0.1	0.50 (0.44)	0.33 (0.61)	0.33 (0.17)

Manhattan distance is positive and the greater distance is, the bigger the identification error is.

The empirical results on the identification error in the parameter space: For each of the eighty one agents in our empirical study, there are nine collections of signatures generated by the specific agent, three collections from each of the three vocabulary sets. For each of these nine collections, we quantitatively measured the identification error (as defined in Equation 4) between the agent and the most likely agent identified by Algorithm 1. Table III, Table IV, and Table V render the average identification errors for all these eighty one agents, corresponding to the eighty one different parameter settings over the four characteristic parameters. For each agent, there are two numbers reported in the tables regarding the average identification errors in the entry associated with the specific parameter setting of the agent. The number inside the parentheses is the average identification error for signature collections from the single-word vocabulary set (JUMPS) while the number outside the parentheses is the average identification error for signature collections from the other two vocabulary sets (RNT and QBF).

V. SUMMARY OF THE EMPIRICAL FINDINGS ABOUT THE ACCURACY OF IDENTITY RECOGNITION

Based on the empirical results depicted in Table II in Section IV above, we have the following empirical findings regarding the accuracy of identity recognition in the virtual circular keyboard domain.

- Given the nine possible combinations regarding the vocabulary set used and the number of signatures in the collection available for identity recognition, (i) with a high accuracy ranging from 90.1% to 95.1% of the

TABLE IV
MEASURING IDENTIFICATION ERROR IN MANHATTAN DISTANCE
(OBSERVATION METRIC LOCALITY $deg_{obs} = 2$)

Transition metric locality: $deg_{tran} = 4$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.22 (0.50)	0.00 (0.33)	0.64 (0.17)
0.4	0.56 (0.67)	0.61 (0.50)	0.14 (0.67)
0.1	0.61 (0.67)	0.39 (0.44)	0.42 (0.50)
Transition metric locality: $deg_{tran} = 3$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.78 (0.56)	0.39 (0.72)	0.50 (0.17)
0.4	0.64 (0.22)	0.58 (0.39)	0.36 (0.67)
0.1	0.86 (0.83)	0.83 (0.56)	0.64 (0.61)
Transition metric locality: $deg_{tran} = 2$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.69 (0.44)	0.39 (0.11)	0.17 (0.67)
0.4	0.28 (0.44)	0.58 (0.33)	0.50 (0.39)
0.1	0.64 (0.61)	0.47 (0.50)	0.53 (0.61)

TABLE V
MEASURING IDENTIFICATION ERROR IN MANHATTAN DISTANCE
(OBSERVATION METRIC LOCALITY $deg_{obs} = 1.5$)

Transition metric locality: $deg_{tran} = 4$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.42 (0.28)	0.14 (0.33)	0.25 (0.33)
0.4	0.22 (0.50)	0.28 (0.17)	0.17 (0.28)
0.1	0.42 (0.33)	0.44 (0.50)	1.03 (0.00)
Transition metric locality: $deg_{tran} = 3$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.58 (0.28)	0.25 (0.00)	0.36 (0.17)
0.4	0.61 (0.44)	0.25 (0.61)	0.69 (0.17)
0.1	0.81 (0.61)	0.56 (0.17)	0.42 (0.28)
Transition metric locality: $deg_{tran} = 2$			
p_{iter}	$p_{mode} = 0.9$	$p_{mode} = 0.6$	$p_{mode} = 0.3$
0.7	0.22 (0.17)	0.45 (0.22)	0.22 (0.17)
0.4	0.58 (0.13)	0.36 (0.11)	0.14 (0.11)
0.1	0.58 (0.94)	0.64 (0.17)	0.47 (0.56)

time, the true agent was correctly recognized as one of the top 5 most likely agents, (ii) with an accuracy ranging from 67.9% to 85.2% of the time, the true agent was correctly recognized as one of the top 3 most likely agents, and (iii) with an accuracy ranging from 27.9% to 49.4% of the time, the true agent was correctly recognized as the most likely agent.

- Given fifty signatures available in the collection for identity recognition, the accuracy of correctly recognizing the true agent as one of the top k most likely agents was consistently better when the vocabulary set is smaller.

Based on the empirical results depicted in Table III to Table V, we have the following empirical findings regarding the error of identity recognition as a normalized Manhattan

distance between the true agent and the most likely agent in the parameter space.

- The average of identification errors (as the distance between the true agent and the most likely agent) over all three tables combined is 0.43 for the multi-word vocabulary sets RNT and QBF and 0.38 for the single-word vocabulary set JUMPS. As a contrast, the average normalized Manhattan distance of two points in our parameter space is $\frac{4}{3}$. This demonstrates the significance of the effectiveness of the identity recognition algorithm for tracking the most likely agent given the collection of signatures.
- The identification errors in Table III (with an average of 0.36 for the vocabulary sets RNT and QBF and also 0.36 for the vocabulary set JUMPS), where the observation metric locality is high with $deg_{obs} = 3$, are consistently lower than the corresponding ones in Table IV (with an average of 0.50 for the vocabulary sets RNT and QBF and 0.49 for the vocabulary set JUMPS), where the observation metric locality is lowered to $deg_{obs} = 2$). However, unlike the results in [4], there are no other clear patterns regarding how certain parameter values affect the identification errors.

VI. CONCLUSIONS AND FUTURE WORK

In this paper, we investigate the identity recognition problem for agents repeatedly engaging in a collection of tasks using resources supported by a common environment. Computer systems and interconnected sensor networks can collect sequences of observations from these agents, which serve as their behavioral signatures. The common environment defines an alternative-paths workflow shared by all the agents. We can model the operational processes of such agents as Hidden Markov models, which share the same state-transition topology defined by the common alternative-paths workflow but have characteristic differences in their specific state transition probabilities and observation probabilities. The goal of the identity recognition problem is to identify the most likely agents given a collection of signatures generated by an unidentified agent.

We propose the modeling framework, define the identity recognition problem, and develop a computational approach for solving the identity recognition problem. In particular, this modeling framework for the identity recognition problem has an interesting application in an virtual-reality environment where people generate text output using a circular virtual keyboard investigated in previous research work [4], [18]. We conduct an empirical study to apply our modeling framework and computational approach for identity recognition in this domain. As illustrated in Section V, our empirical findings demonstrate the fundamental effectiveness of the approach in tracking the most likely agents given a collection of signatures. The accuracy of identity recognition could be further improved if the conditional

probability $Pr(W|P)$ were parameterized differently for different agent P , rather than being the non-distinguishing uniform distribution assumed in Assumption 1. As future work, we plan to further investigate how the presence of such distinguishing probability distributions affects the accuracy of identity recognition.

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